SMOOTHNESS AND ETALENESS NEAR A NOETHERIAN CLOSED SUBSPACE ARE DETECTED ON ITS INFINITESIMAL NEIGHBOURHOODS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 5230-5243 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 5245-5256, the complete author proof (10 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Counted B side (spaces, more general; both sides complete).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Chapter lemma lifting-along-artinian-at-point; the infinitesimal factorization through Artinian local rings; etale-vs-smooth-dimension-zero dictionary.

Source reference. Original lemma 02HX: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 5096–5116. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Assume f is locally of finite type and Y locally Noetherian. Let $Z \subset Y$ be a closed subspace with nth infinitesimal neighbourhood $Z_n \subset Y$. Set $X_n = Z_n \times_Y X$.*

(1) *If $X_n \to Z_n$ is smooth for all n, then f is smooth at every point of $f^{-1}(Z)$.*
(2) *If $X_n \to Z_n$ is étale for all n, then f is étale at every point of $f^{-1}(Z)$.*

Proof. Assume $X_n \to Z_n$ is smooth for all n. Let $x \in X$ be a point lying over a point of Z. Given a small extension $B' \to B$ and morphisms α, β as in Lemma 02HX part (3) the maximal ideal of B' is nilpotent (as B' is Artinian) and hence the morphism β factors through Z_n and α factors through X_n for a suitable n. Thus the lifting property for $X_n \to Z_n$ kicks in to get the desired dotted arrow in the diagram. This proves (1). Part (2) follows from (1) and the fact that a morphism is étale if and only if it is smooth of relative dimension 0. □

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.